

Hotel Booking Analytics Assessment

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Section 1: Data Quality Checks

C1. Bookings with num_rooms = 0

Analysis

A total of 60 bookings (0.50% of all bookings) have num_rooms = 0.

Interpretation

A booking with zero rooms is generally not a valid business transaction and likely indicates a data quality issue in the upstream pipeline. Potential causes include incomplete booking creation, failed validation checks during data entry, or ETL processes that allowed records to be ingested before room allocation information was populated.

C2. Booking Date Earlier than Customer Signup Date

Analysis

A total of 163 bookings (1.36% of all bookings) have a booking date earlier than the customer signup date.

Interpretation

This pattern does not necessarily indicate data corruption.

Possible explanations include:

1. Customers may have made bookings as guest users and created accounts later.
2. Delays in account creation events, ETL synchronization, or timezone differences may result in signup timestamps appearing later than booking timestamps.

While the volume is relatively small, these records should be reviewed to confirm whether they represent expected business behavior.

C3. Missing Reviews Analysis

Analysis

A total of 6,854 bookings (57.12%) has no review recorded.

Breakdown by Booking Status

Booking Status	Missing Review %
Completed	45.40%
Cancelled	97.83%
No-Show	100.00%

Interpretation

The pattern is largely consistent with expected business behavior.

- No-Show bookings have a 100% missing review rate because the stay never occurred.
- Cancelled bookings show a 97.83% missing review rate because customers typically do not review cancelled stays.
- Completed bookings exhibit a 45.40% missing review rate, which may indicate low review participation or gaps in review collection processes.

Insert Visualization: Missing Reviews by Booking Status

Section 2: VIP vs Volume Strategy

C1. Define VIP Three Ways

VIP customers were defined using three independent criteria calculated using completed bookings only:

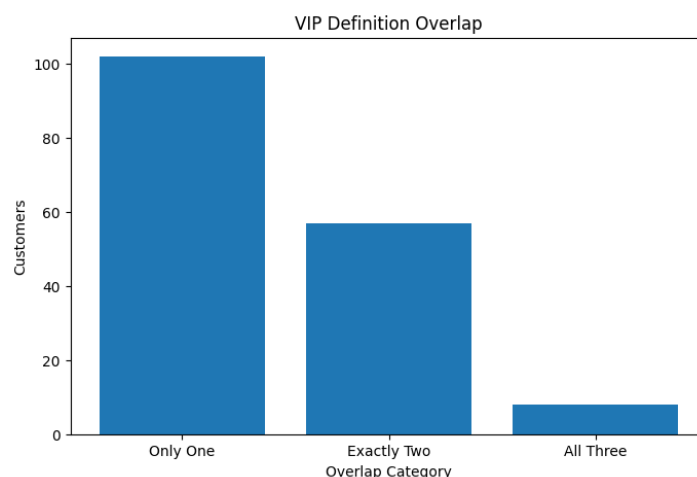
1. Top 10% by Total Spend
2. Top 10% by Booking Count
3. Top 10% by Lifetime Value Rate (Total Spend ÷ Months Since Signup)

Overlap Results

Category	Customers
Only One Definition	102
Exactly Two Definitions	57
All Three Definitions	8

Interpretation

Only 8 customers appear in all three VIP definitions, indicating that VIP status is highly sensitive to the chosen definition. High-spending customers are not necessarily the most frequent bookers, and neither group perfectly overlaps with customers generating the highest spend relative to tenure.



Visualization: VIP Definition Overlap

C2. VIP Risk Profile

Selected VIP Definition

The spend-based VIP definition was selected because revenue contribution is the most directly aligned metric with business value.

Revenue Share

VIP customers contribute **26.84%** of realized platform revenue.

Cancellation Rate

Group	Cancellation Rate
VIP	17.06%
Non-VIP	19.45%

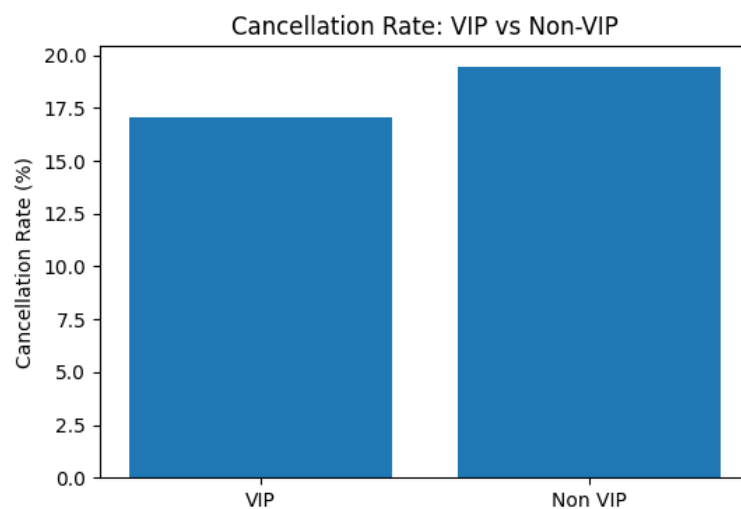
Discount Intensity

Group	Average Discount
VIP	₹3,967.06
Non-VIP	₹1,439.99

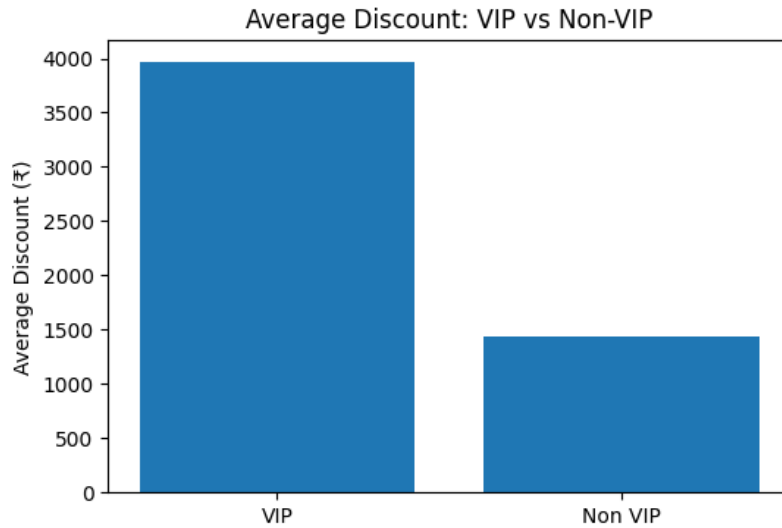
Interpretation

VIP customers exhibit lower cancellation rates than non-VIPs, suggesting stronger engagement and lower behavioral risk.

However, VIP customers currently receive approximately 2.75x more discounts than non-VIPs despite demonstrating lower cancellation risk.



Visualization: VIP vs Non-VIP Cancellation Rate



Visualization: VIP vs Non-VIP Discount Intensity

C3. Acquisition Economics

Assumptions

Budget = ₹50,00,000

Customer Acquisition Cost (CAC) = ₹3,000

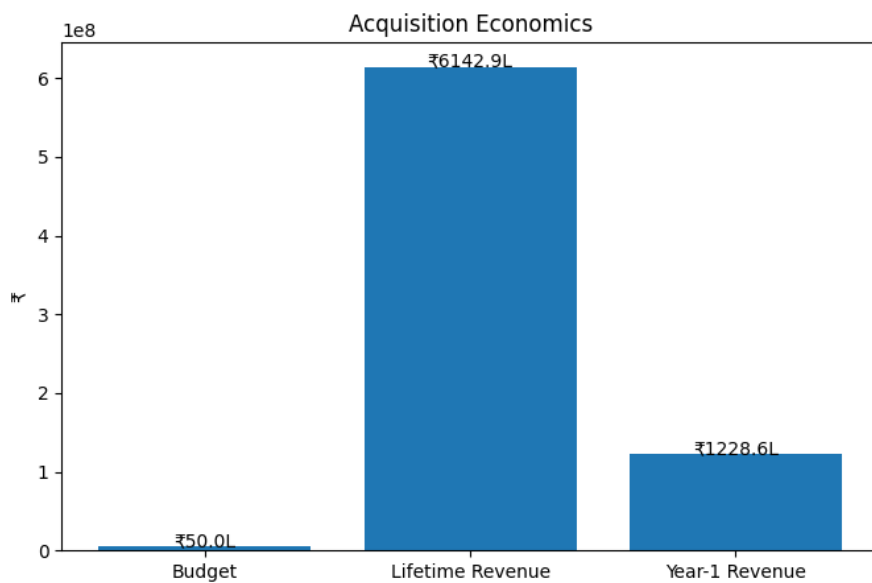
Results

New Customers Acquired: 1,667 customers

Average Customer Lifetime Spend: ₹368,574.74

Gross Lifetime Revenue: ₹614,291,232.52

Year-1 Revenue (20% realization): ₹122,858,246.50



Visualization: Acquisition Economics

C4. Retention Economics

Assumptions

Retention Cost per Customer = ₹500

Results

Customers Retained:

10,000 customers

Break-Even Lifetime Spend:

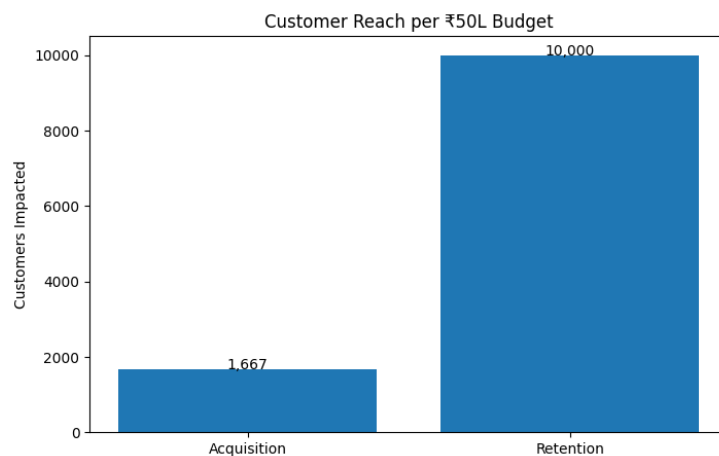
₹2,500

Actual Average Customer Lifetime Spend:

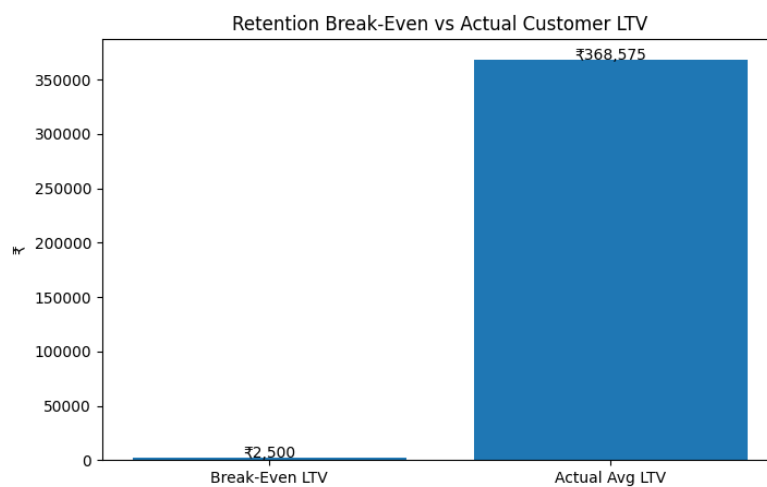
₹368,574.74

Interpretation

The actual customer lifetime spend is substantially higher than the break-even threshold, indicating that retaining valuable customers can be economically attractive when targeted appropriately.



Visualization: Customer Reach per ₹50L Budget



Visualization: Break-Even LTV vs Actual Customer LTV

Recommendation

Based on the analysis, a balanced allocation strategy is recommended.

VIP customers contribute a disproportionate share of platform revenue while exhibiting lower cancellation rates. Retention economics are highly favorable because average customer lifetime spend significantly exceeds the break-even threshold.

However, acquisition remains attractive due to the substantial projected lifetime revenue generated by newly acquired customers.

Recommended budget split:

- 60% Retention
- 40% Acquisition

This allocation prioritizes the protection of high-value customers while continuing to expand the customer base.

Section 3: SQL Challenge

Schema Design

Customers Table

```
CREATE TABLE customers (  
  customer_id INT PRIMARY KEY,  
  customer_name VARCHAR(100) NOT NULL,  
  customer_segment VARCHAR(30),  
  customer_signup_date DATE NOT NULL,  
  customer_home_city VARCHAR(100),  
  customer_loyalty_tier VARCHAR(20)  
);
```

Properties Table

```
CREATE TABLE properties (  
  property_id INT PRIMARY KEY,  
  property_name VARCHAR(150) NOT NULL,  
  property_city VARCHAR(100) NOT NULL,  
  property_star_rating DECIMAL(2,1),  
  property_type VARCHAR(50),  
  property_total_rooms INT NOT NULL,  
  
  CONSTRAINT uq_property  
  UNIQUE(property_name, property_city)  
);
```

Bookings Table

```
CREATE TABLE bookings (  
  booking_id INT PRIMARY KEY,  
  customer_id INT NOT NULL,  
  property_id INT NOT NULL,  
  booking_date DATE NOT NULL,  
  checkin_date DATE NOT NULL,  
  checkout_date DATE NOT NULL,  
  room_type VARCHAR(50),  
  num_rooms INT NOT NULL,
```

```
    nights INT NOT NULL,  
    booking_channel VARCHAR(30),  
    adr DECIMAL(10,2),  
    discount_amount DECIMAL(12,2),  
    coupon_code VARCHAR(50),  
    total_amount DECIMAL(12,2) NOT NULL,  
    payment_method VARCHAR(50),  
    booking_status VARCHAR(20) NOT NULL,
```

```
    FOREIGN KEY (customer_id)  
        REFERENCES customers(customer_id),
```

```
    FOREIGN KEY (property_id)  
        REFERENCES properties(property_id),
```

```
    CHECK (checkout_date > checkin_date)
```

```
);
```

Reviews Table

```
CREATE TABLE reviews (  
    booking_id INT PRIMARY KEY,  
    review_rating VARCHAR(20),  
    review_date DATE,
```

```
    FOREIGN KEY (booking_id)  
        REFERENCES bookings(booking_id)
```

```
);
```

Index Recommendation

```
CREATE INDEX idx_bookings_customer  
ON bookings(customer_id);
```

Justification

Customer-level analysis drives retention, VIP segmentation, booking sequence analysis, and lifetime value calculations. Indexing customer_id improves join performance and accelerates analytical queries.

SQL Query C-Q1

Query

```
WITH ranked_bookings AS (  
  
    SELECT  
        customer_id,  
        total_amount,  
  
        ROW_NUMBER() OVER (  
            PARTITION BY customer_id  
            ORDER BY checkin_date  
        ) AS booking_sequence  
  
    FROM hotel_bookings  
    WHERE booking_status = 'Completed'
```

```
)
```

```
SELECT
```

```
    customer_id,  
    total_amount,
```

```
    ROW_NUMBER() OVER (  
        PARTITION BY customer_id  
        ORDER BY checkin_date  
    ) AS booking_sequence
```

```
FROM hotel_bookings  
WHERE booking_status = 'Completed'
```

```
)
```

```
SELECT
```

```

CASE
  WHEN booking_sequence = 1 THEN 'First'
  WHEN booking_sequence = 2 THEN 'Second'
  ELSE 'Third+'
END AS sequence_label,

COUNT(*) AS n,
ROUND(AVG(total_amount), 2) AS avg_total

```

```
FROM ranked_bookings
```

```

GROUP BY
CASE
  WHEN booking_sequence = 1 THEN 'First'
  WHEN booking_sequence = 2 THEN 'Second'
  ELSE 'Third+'
END;

```

Results

Sequence	N	Avg Total
First	800	₹33,613.74
Second	799	₹30,863.72
Third+	7,734	₹31,459.62

Interpretation

First-time bookings exhibit the highest average booking value, while repeat bookings stabilize at a slightly lower but consistent level.

SQL Query C-Q2

Query

```

WITH normalized_reviews AS (

SELECT
  property_name,
  property_city,

CASE
  WHEN booking_channel = 'Corporate Portal'
  THEN CAST(review_rating AS FLOAT) / 2.0
  ELSE CAST(review_rating AS FLOAT)
END AS normalized_rating

FROM hotel_bookings

WHERE review_rating IS NOT NULL
)

SELECT TOP 5
  property_name,
  property_city,

```

```

COUNT(*) AS num_reviews,

CAST(
AVG(normalized_rating)
AS DECIMAL(5,2)
) AS normalized_avg_rating

FROM normalized_reviews

GROUP BY
property_name,
property_city

HAVING COUNT(*) >= 20

ORDER BY normalized_avg_rating DESC;

```

Results

Property	City	Reviews	Avg Rating
Indigo Stay	Mumbai	96	4.35
Banyan Retreat	Bangalore	80	4.29
Olive Courtyard	Mumbai	101	4.29
Sea Breeze Resort	Goa	78	4.26
Saffron Residency	Pune	100	4.19

Interpretation

Indigo Stay emerged as the highest-rated property after normalization and minimum-review filtering. Applying a minimum review threshold improves ranking reliability by reducing noise from small sample sizes.